Agrim Srivastava

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Education

Indian Institute of Technology Madras Sept 2023 – Sept 2027
BS in Data Science and Applications **GPA: 9.1 / 10.0**

Experience

Software Development Engineer Intern, annam.ai Jan 2026 – Mar 2026

- Resolved critical issues in **authentication & RBAC flows**, improving system security
- Built **time based question generation** from live transcripts enabling real time interaction
- Developed dashboards using **React (Vite, Zustand, React Query, TanStack Router)**
- Contributed to backend services using **Node.js, Express, MongoDB**; integrated **Firebase Admin**
- Collaborated via **Git workflows**, shipping multiple production ready features

Full-Stack Software Developer — Client Project, TutorSolve Month Year – Present

- Worked directly with a client to **gather requirements, design, develop, and deploy** a production full-stack marketplace now serving **users across multiple countries**
- Architected a **4-role marketplace** (student, expert, employee-admin, super-admin) using **Flask REST APIs, MongoDB Atlas, and JWT-based RBAC**, with geo fencing to enforce region specific access policies
- Designed and implemented the **transaction engine**, a multi-phase order lifecycle covering expert matching, Maker-Checker price approval, split-payment (advance + completion) via Stripe, and file delivery with a **25% preview lock** enforced until full payment
- Built a **real time dual channel chat system** using Flask-SocketIO and eventlet, running isolated Socket.IO rooms per thread (student-admin and expert-admin), replacing a polling model with zero latency message delivery
- Engineered a **KYC verification workflow** with multifile upload, admin approval gates, and domain filtered job board access, ensuring experts see only questions within their registered specialisation
- Delivered **five expert facing dashboard pages** and a full admin cockpit with split view chat, pricing controls, assignment search, file relay, and a rolling average quality score engine

Teaching Assistant, IIT Madras Oct 2025 – Sept 2026

- Supported **Tools in Data Science (TDS)** through hands-on sessions covering data processing, analysis, visualization, APIs, automation, and LLM-based workflows
- Designed and developed **practical coding problems and exercises** focused on applying Python and modern data science tools to real-world workflows
- Conducted technical sessions on **probability, statistics, and data science workflows**, translating concepts into implementation-focused examples
- Resolved technical queries on Discourse and guided students through **debugging, implementation, and tool usage**
- Evaluated assignments with focus on **correctness, implementation quality, and problem-solving approach**

Projects

Household Services App

- Built fullstack platform using **Flask (REST APIs)** and **Vue.js** supporting multirole access
- Designed **RBAC system** ensuring secure authentication and data validation
- Implemented **asynchronous processing** using Redis and Celery
- Designed normalized **SQLite schema** for scalable data management
- Developed dashboards using **Vue.js + Pinia** for state management

RAG based Teaching Assistant

- Built end-to-end **RAG system** for answering IITM course queries
- Developed backend using **FastAPI**; integrated **Ollama (embeddings)** and **Gemini (multimodal)**
- Engineered **parallel embedding pipeline** improving processing efficiency
- Implemented **cosine similarity search + caching** with compressed storage
- Built interface supporting **text + image queries**

Coursework

Machine Learning: Foundations, Techniques, Practice, Deep Learning
Computer Science: DSA, DBMS, System Commands
Development: Modern Application Development, Software Testing
Humanities: Strategies for Professional Growth

Skills

Languages: Python, JavaScript
Backend: FastAPI, Flask, Node.js
Frontend: Vue.js, React, Bootstrap
Databases: PostgreSQL, SQLite, MongoDB
Tools: Redis, Celery, Git, Linux